

MTH441: Lab 8

Analysis of Variance

1. Consider the `PlantGrowth` dataset in `R` that contains results from an experiment comparing yields of a plant under a control group and two treatments. First, let's visualize the data:

```
data("PlantGrowth")
boxplot(weight ~ group, data = PlantGrowth)
```

Using the boxplot created, can you guess whether there is a significant difference between the mean yield under each of the three levels?

2. Using no in-built functions, calculate the sums of squares of treatment and errors and carry out the F test to test for difference of means for this dataset. Do you reject the null hypothesis?
3. You can use the `aov` function to carry out the ANOVA test:

```
one_way_aov <- aov(weight ~ group, data = PlantGrowth)
summary(one_way_aov)
```

Make sure the analysis matches with Question 2.

4. Repeat the above for the `InspectSprays` dataset with an appropriate ANOVA test.
5. A fast-food franchise is test-marketing 3 new menu items. To find out if they have the same popularity, 6 franchisee restaurants are randomly chosen for participation in the study. In accordance with the **randomized block design**, each restaurant will test market all 3 new menu items. The data in Table 2 represent the sales figures of the 3 new menu items in different restaurants. Perform the analysis of variance to test the equality of all three items.

Table 2: Restaurant Experiment

Restaurant	Item1	Item2	Item3
R1	31	27	24
R2	31	28	31
R3	45	29	46

Restaurant	Item1	Item2	Item3
R4	21	18	48
R5	42	36	46
R6	32	17	40

6. For the `ToothGrowth` dataset, there are two treatment levels. Using the code below, can you assess whether there an interaction effect for the two variables:

```
means <- aggregate(len ~ dose + supp, data = ToothGrowth, FUN = mean)

# Plot with lines for each supplement
plot(len ~ dose, data = subset(means, supp == "OJ"),
     type = "b", pch = 19, col = "orange", ylim = c(0, max(means$len)),
     xlab = "Dose (mg/day)", ylab = "Tooth length",
     main = "Tooth growth by dose and supplement")

lines(len ~ dose, data = subset(means, supp == "VC"),
     type = "b", pch = 17, col = "blue")

legend("topleft", legend = c("OJ", "VC"),
     col = c("orange", "blue"), pch = c(19, 17), lty = 1)
```

7. Implement a two-way ANOVA model for the `ToothGrowth` dataset and conclude which supplement is better and how dose affects the choice of supplements.